Learning-based Speculator (§3)

[inputs]

Ensembled Small Speculative Models (SSMs)

Fine-tuned SSM

Quantized LLM

Distillated LLM

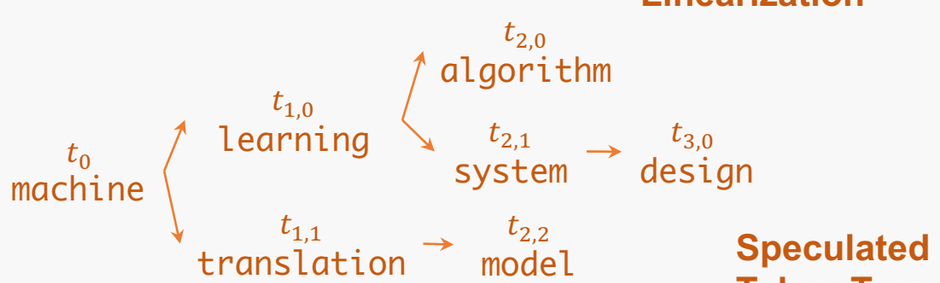
Knowledge Retriever

User-Defined Function

Verified output: machine learning system optimization

Output 0: [machine] learning algorithm
Output 1: [machine] learning system design
Output 2: [machine] translation model

Linearization

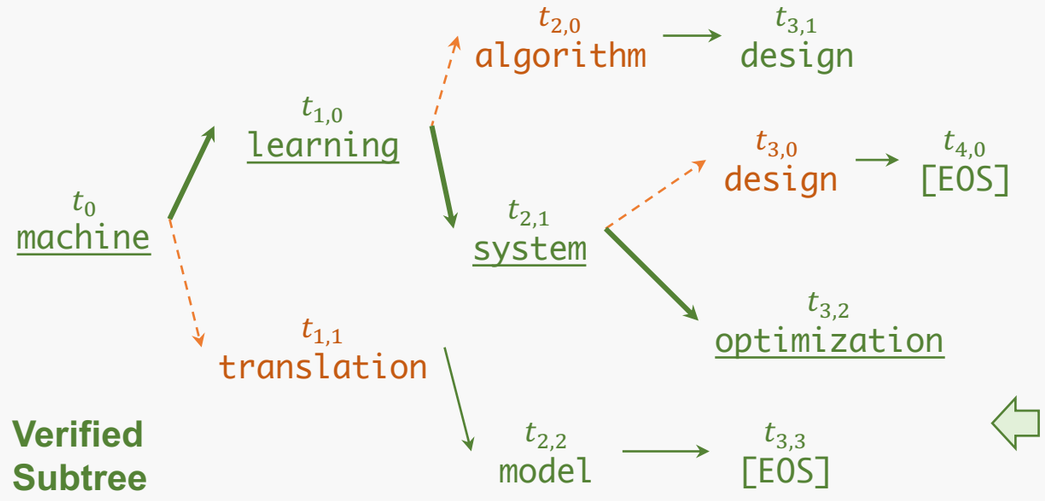


Speculated Token Tree

Synchronization

Linearization

Token Tree Verifier (§4)



Verified Subtree

[machine]→learning→algorithm→system→design→translation→model

Tree-based Parallel Decoding

LLM

learning system design optimization [EOS] model [EOS]